

WEEK 10

SAR & Sensor Fusion

Piercing the Clouds: All-Weather Intelligence for ARIA v7.0

全天候監測與資料融合

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THE CAPTAIN'S PROBLEM: THE WHITE WALL

雲層遮蔽 — 光學衛星的致命弱點

OPTICAL (Sentinel-2)

✗ During typhoon:

Cloud cover ~90%

- WHITE IMAGE
- Zero ground info
- Blind to flooding

SAR (Sentinel-1)

✓ During typhoon:

Microwaves penetrate clouds

- CLEAR IMAGE
- Ground features visible
- Flood extent mapped

90% of critical disaster moments are covered by clouds. SAR is the solution.

WHAT IS SAR? THE ACTIVE EYES

合成孔徑雷達：主動感測的眼睛

Optical = Passive

- Waits for sunlight
- Measures reflected visible/IR
- Blocked by clouds & darkness
- Rich spectral info (color)

SAR = Active

- Emits own microwave pulses
- Measures backscatter (σ^0)
- Penetrates clouds & works at night
- Measures roughness & structure

Why microwaves pass through clouds: C-band wavelength (~5.6 cm) is ~5000× larger than cloud droplets (~10 μm). Microwaves simply pass through — like your hand through fog.

BACKSCATTER (後向散射)

SAR measures roughness, not color

Surface	Mechanism	Typical dB	SAR Image
Calm Water	Specular (mirror)	< -20 dB	Very Dark
Rough Water	Diffuse scattering	-15 to -10 dB	Dark–Medium
Bare Soil	Surface scattering	-10 to -5 dB	Medium
Forest	Volume scattering	-8 to -3 dB	Bright
Buildings	Double-bounce	> 0 dB	Very Bright

Why negative? $\text{dB} = 10 \times \log_{10}(\sigma^0)$ $\sigma^0 < 1$ (returned < transmitted) $\rightarrow \log < 0 \rightarrow \text{dB is negative. e.g. } \sigma^0 = 0.01 \rightarrow -20 \text{ dB}$

SPECULAR REFLECTION: WHY WATER IS BLACK

鏡面反射：水面為什麼是黑的？

The Mirror Analogy

Smooth water acts like a mirror.

Radar pulse hits the surface → reflects AWAY at the same angle.

Almost no energy returns to the satellite.

Result: Water appears BLACK (very low backscatter)

ARIA v7.0 Flood Indicator

$VV < -18 \text{ dB} \rightarrow \text{Water}$

This single threshold is our key flood detection rule.

⚠ Caveat: Rough water (wind waves) breaks specular reflection → backscatter increases → may exceed -18 dB threshold → missed detection.

SENTINEL-1: THE SAR WORKHORSE

C-band SAR 全球監測衛星



C-band (5.6 cm)



~10 m (IW GRD)



6 days (2 satellites)



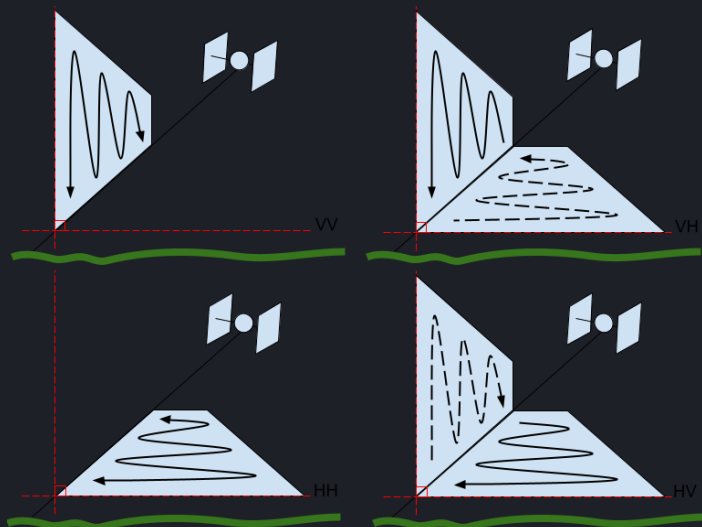
VV (vertical struct. / water) + VH (canopy / flooded veg.)

Sentinel-1 User Guide → <https://sentinels.copernicus.eu/web/sentinel/user-guides/sentinel-1-sar>

VV vs VH POLARIZATION

VV 與 VH 極化：地表的兩種觀察角度

Polarization = pulse orientation. **VV** = send V, receive V. **VH** = send V, receive H (cross-pol).



VV (同極化 co-pol)

- **Water:** < -20 dB — specular reflection, almost no return
- **Vertical structures:** > -5 dB — buildings, standing trees

水陸對比極大 → 洪水開放水面偵測首選

ARIA v7.0: VV < -18 dB → water mask

VH (交叉極化 cross-pol)

- **Dense canopy:** -8 to -3 dB — volume scattering (V→H)
- **Flooded veg:** double-bounce (water surface + trunk)

樹枝把 V 脈衝散射轉換方向 (V→H) · 植被越密 VH 越強。

Application: Forest density, crop, flooded vegetation

W10: 本週用 VV 做開放水面偵測。參考圖解 → <https://gis.ardswc.gov.tw/themes/satellite/41> | <https://esthub.maarum.ee/eng/help/esthub-data/sentinel-1-p89.html>

SAR GEOMETRIC LIMITATIONS

SAR 幾何限制 — 地形與軌道的影響

Foreshortening & Layover (前縮 / 疊置效應)

Slopes facing radar are compressed (foreshortening) or folded over (layover) → false dark patches → misidentified as water

Fix: DEM slope filter (W4): slope > 15° → not water

Radar Shadow (雷達陰影 / 遮蔽效應)

Steep slopes AWAY from the radar receive no signal → zero backscatter → dark zone looks like water but is actually a mountain shadow

Fix: Use DEM aspect + orbit direction to flag shadow zones

Ascending vs Descending Orbit (升軌 vs 降軌)

Ascending (northbound, dusk) and descending (southbound, dawn) orbits view slopes from opposite sides → different distortions on the same terrain

Fix: Compare both passes; shadow on one orbit may be visible on the other

Incidence Angle (入射角效應)

Near-range (steep angle ~30°) exaggerates distortion; far-range (shallow ~45°) reduces it. Backscatter intensity also varies with angle across the swath

Fix: Apply angle normalization; be aware of cross-swath variation

Live Demo → <https://search.asf.alaska.edu> | ESA SAR Tutorial → <https://earth.esa.int/eogateway/learn/sar>

SAR SIGNAL LIMITATIONS

SAR 訊號限制 — 雜訊與光譜盲區

Speckle Noise (斑點雜訊)

Coherent wave interference creates salt-and-pepper pattern → false water pixels everywhere

Fix: Apply Median Filter (5×5) before any thresholding

Wind Roughening (風浪干擾)

Strong winds break the smooth water surface → backscatter rises above -18 dB → flood missed

Fix: Cross-check with optical NDWI (sensor fusion, ARIA v7.0)

No Spectral Info (無光譜資訊)

SAR sees roughness only — no color, no NDVI, no vegetation type discrimination

Fix: Fuse with optical (Sentinel-2) for spectral discrimination

重點： 每一個限制都有對策 — 這就是 ARIA 多模組系統設計的意義。

SPECKLE: THE SALT & PEPPER NOISE

斑點雜訊：SAR 的固有問題

The Problem

SAR uses coherent waves.
Constructive + destructive interference creates a grainy pattern.

Thresholding raw SAR → thousands of false water pixels!

The Solution: Median Filter

Apply a spatial filter (e.g., 5×5 median) that smooths noise while preserving edges.

Pipeline:

Raw SAR → Median Filter → Threshold

Never skip this step!

Captain's Audit: Never run a threshold on raw SAR; filter it first!

LAB 1

SAR FLOOD DETECTION

SAR 淹水偵測 | 35 min

1

Load S1_Hualien_dB.tif

Pre-processed Sentinel-1 VV in dB

2

Threshold: $\text{sar} < -18 \text{ dB}$

Extract water pixels via backscatter threshold

3

Median Filter (5×5)

Remove speckle noise before final classification

Key Question: *What did the radar see that the optical satellite missed during the typhoon?*

SENSOR FUSION: THE ARIA v7.0 SECRET SAUCE

資料融合：v7.0 的核心能力

Optical Strengths & Limits

- ✓ Rich spectral detail (NDVI, NDWI)
- ✓ Color discrimination
- ✗ Cloud-blind
- ✗ Night-blind

SAR Strengths & Limits

- ✓ Cloud-piercing (all-weather)
- ✓ Day & night operation
- ✗ Speckle noise
- ✗ No color info (no NDVI)

The Hybrid Approach

Neither sensor is complete alone. Combining both produces a HIGH-CONFIDENCE map:

Optical confirms spectral identity + SAR confirms physical presence = dual evidence.

This is the core principle of ARIA v7.0.

FUSION LOGIC: THE CONFIDENCE MATRIX

融合邏輯：確信度矩陣

	Optical	SAR	Result
HIGH CONFIDENCE	NDWI > 0.3	VV < -18	Both sensors agree
SAR ONLY (CLOUDY)	Cloudy	VV < -18	Radar pierces clouds
OPTICAL ONLY	NDWI > 0.3	VV ≥ -18	Needs manual review
NO DETECTION	No signal	VV ≥ -18	No flood evidence

v7.0 fills the optical cloud gaps with SAR data → complete operational picture.

MULTI-SOURCE DATA FUSION

ARIA v7.0 資料融合 | 30 min

1

Apply Fusion Logic

High Conf: (NDWI > 0.3) & (SAR < -18) | SAR Only: (Cloud) & (SAR < -18)

2

Topographic Correction

Use W4 slope map: slope > 15° → false positive (water can't pool on cliffs)

3

AI Strategic Briefing

Feed metrics to Gemini → generate an emergency response directive for Hualien

Output: *ARIA v7.0 Consolidated Flood Map — integrating Optical + SAR + DEM*

TOPOGRAPHIC FILTER: USING W4 INTELLIGENCE

地形審計：利用 W4 坡度修正 SAR 誤報

The Problem

SAR on steep slopes suffers from:

Foreshortening (前縮效應)

Slopes facing radar appear compressed

Layover (疊置效應)

Very steep slopes cause signal overlap

Both create false low backscatter → misidentified as water!

The Fix

Rule:

If flood detected BUT slope > 15°

→ **Reclassify as FALSE POSITIVE**

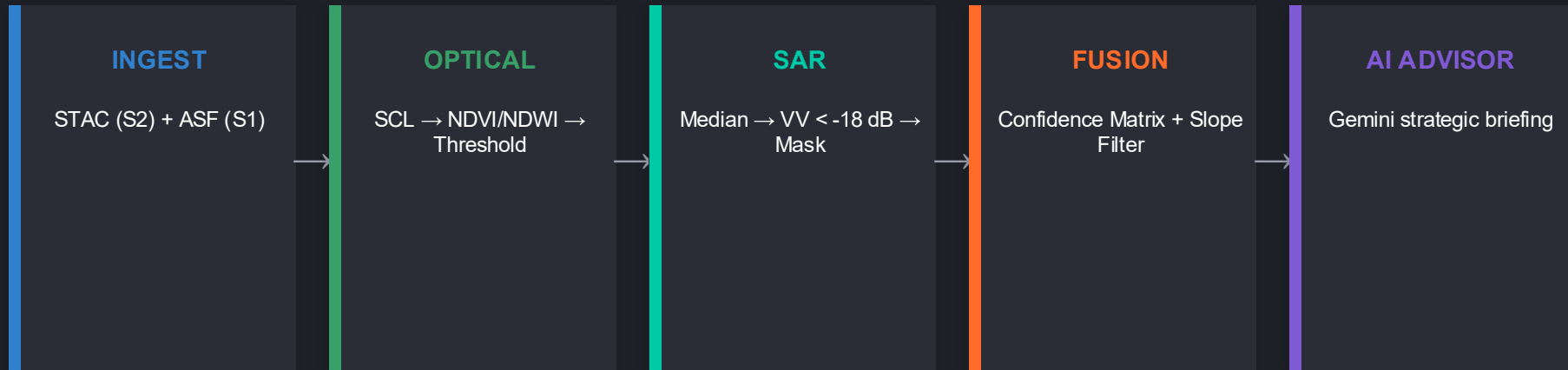
Physical logic:

Water cannot pool on a cliff.

This uses the DEM slope raster from Week 4 — earlier ARIA modules feed into later ones.

ARIA v7.0 ARCHITECTURE: ALL-WEATHER READINESS

系統架構：全天候決策引擎



Output: v7.0 Consolidated Confidence Map + AI Strategic Briefing

MIDTERM SHOWCASE PREVIEW (W11)

期中成果發表預告

Format

5 min presentation + 2 min Q&A per group

Focus

ARIA v1.0 → v7.0 evolution & cloud-piercing demo

Key Demo

Show how your system detects clouds and switches to SAR

Not About

Coding competition — it's about SYSTEM THINKING

WEEK 10 REFLECTION

"Light is beautiful, but Radio is reliable."

「光很美，但無線電才可靠。」

You are now an All-Weather Intelligence Officer.

你現在是全天候情報官了。

"A commander doesn't care if it's cloudy. He needs the truth. ARIA v7.0 delivers it."

YOUR RESPONSIBILITY: VERIFY BEFORE YOU SUBMIT

作業三原則：跑出來 ≠ 跑對了

1

Filter BEFORE Threshold

Raw SAR + threshold = garbage. Did you apply the Median Filter?

先濾波再分割

2

Apply Topographic Correction

Water on a 40° slope? That's a bug, not a flood.

地形修正不能省

3

Ask, Don't Guess

NTUCool, office hours, or in class. Questions are welcome.

不懂就問，不要猜

An unverified disaster map is more dangerous than no map at all.

未經驗證的災害圖比沒有圖更危險。